

Evaluation Board for Fractional-N PLL Frequency Synthesizer

FEATURES

Self-contained board including synthesizer, VCO, 10 MHz TCXO reference, USB interface, voltage regulators, and loop filter (5.8 GHz)

Designed for 10 MHz PFD frequency, minimum charge pump current, and 20 kHz loop bandwidth

Accompanying software allows control of synthesizer functions from a PC (via USB interface)

Battery operated: choice of 3 V or 5 V supplies

GENERAL DESCRIPTION

The EVAL-ADF4158EB1Z evaluates the performance of the ADF4158 frequency synthesizer for phase locked loops (PLL). A digital picture is shown in Figure 1. It contains the ADF4158 synthesizer, a PC connector, SMA connectors for the power supplies, and an RF output. There is also a low pass loop filter (20 kHz) and a VCO (Z Communications, Inc., VCO940ME03

5.8 GHz) on board. The evaluation board is set up for a 10 MHz PFD comparison frequency. An on-board TCXO provides the 10 MHz reference frequency. A USB cable is included with the board to connect to a PC USB port.

The package also contains Windows® software (XP, Vista, and 7 compatible) to allow easy programming of the synthesizer.

EVALUATION KIT CONTENTS

Antistatic bag with USB warning label (free issue by Analog Devices, Inc.)

Fractional-N synthesizer CD (with USB control) (issue by Analog Devices)

Evaluation board box (Europaks K-645/1) with EVAL-ADF4158EB1Z barcode label

Wrapping material (antistatic bubble wrap)

DIGITAL PICTURE OF THE EVALUATION BOARD



Figure 1.

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REVISION HISTORY

10/12—Rev. 0 to Rev. A

Changes to General Description Section	1
Changes to Evaluation Board Software Section	5
Added Figure 7; Renumbered Sequentially	8

5/10—Revision 0: Initial Version

EVALUATION BOARD HARDWARE

OVERVIEW

The EVAL-ADF4158EB1Z is supplied with a cable for connecting to the USB port of a PC. The silk screen for the evaluation board is shown in Figure 2. The board schematic is shown in Figure 8, Figure 9, and Figure 10.

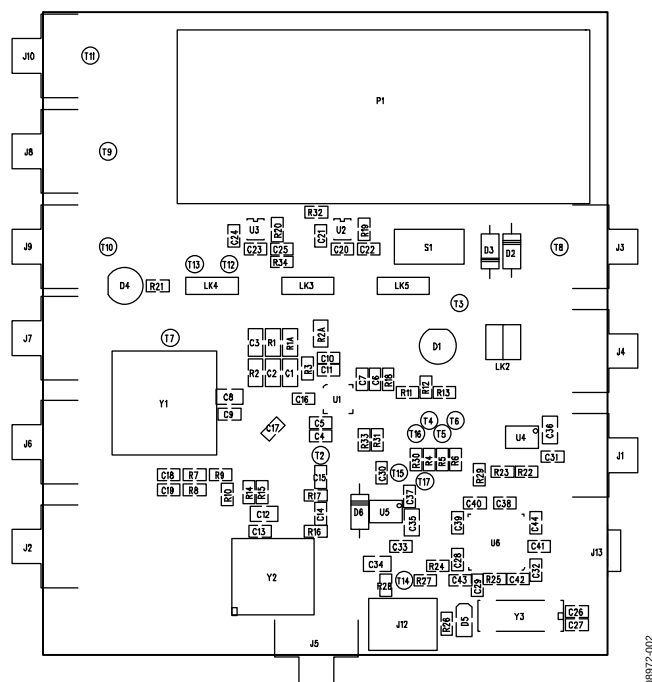


Figure 2. Evaluation Board Silk Screen

POWER SUPPLIES

The board is powered from a single 9 V battery. The power supply circuitry allows the user to choose 3 V for the AV_{DD} , DV_{DD} , and SDV_{DD} pins of the ADF4158 and either 3 V or 5 V for the V_P pin of the ADF4158 and for the VCO supply. The default settings are 3 V for AV_{DD} , DV_{DD} , and SDV_{DD} and 5 V for V_P .

It is very important to note that the AV_{DD} , DV_{DD} , and SDV_{DD} pins of the ADF4158 should never exceed the V_P pin of the ADF4158. This can cause damage to the device.

If the user wishes, external power supplies can be used. In this case, insert SMA connectors as shown on the silk screen (see Figure 2) and the digital picture (see Figure 1) of the evaluation board.

LOOP FILTER

The loop filter schematic is included in the board schematic in Figure 8, Figure 9, and Figure 10. The loop filter component placements are clarified in Figure 3.

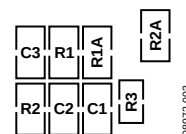


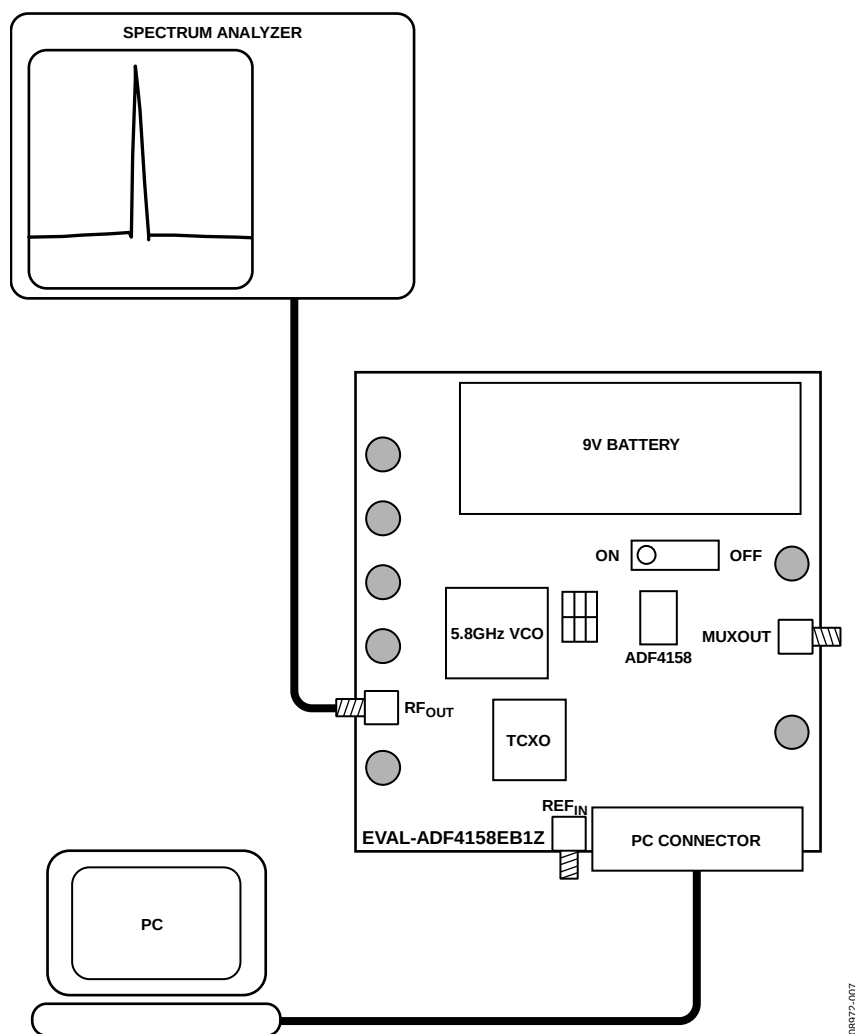
Figure 3. Loop Filter Placement

LOCAL OSCILLATOR COMPONENTS

The 10 MHz TCXO provides the reference frequency. The on-chip R divider should be set to 1 so that the PFD frequency is also 10 MHz. The PLL is made up of the 10 MHz TCXO, the ADF4158, a passive loop filter (20 kHz bandwidth), and the VCO V940ME03 VCO from Z Communications. The output is available at RF_{OUT} through a standard SMA connector.

The on-board loop filter has been designed for a charge pump current of 0.3125 mA. ADIsimPLL, a free download available at www.analog.com/pll, can be used to design other loop filters.

EVALUATION SETUP



08972-007

Figure 4. Evaluation Setup

EVALUATION BOARD SOFTWARE

The control software for EVAL-ADF4158EB1Z is on the CD that accompanies the board. For the software installation procedure, see [UG-476](#).

SELECT DEVICE AND CONNECTION

In the **Select Device and Connection** tab (see Figure 5), choose **ADF4158**, choose **USB board (green)**, and click **Connect**. The software connects to the evaluation board.

MAIN CONTROLS

In the **Main Controls** tab (see Figure 6), the basic PLL settings can be selected. When a register value is changed, the relevant register box at the bottom of the screen turns green. To write

the value to the evaluation board, click the associated button. The box returns to a grey color once the value has been written.

See the **RF Synthesizer: A Worked Example** section of the [ADF4158](#) data sheet for more information.

RAMPS AND SHIFT-KEYING

In the **Ramps and Shift-keying** tab (see Figure 7), ramp parameters can be set and then programmed to the evaluation board.

See the **FMCW Radar Ramp Settings Worked Example** section of the [ADF4158](#) data sheet for more information.

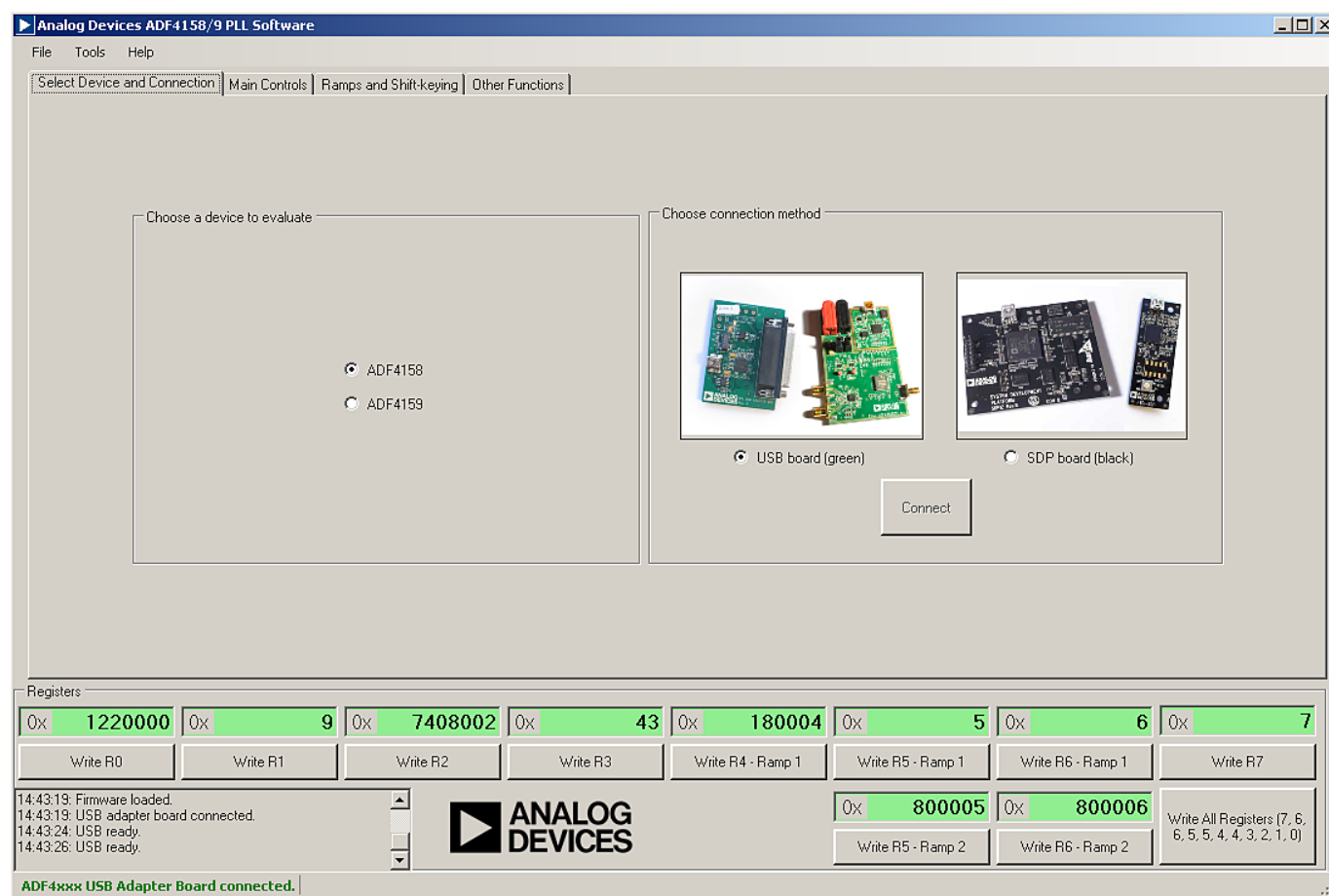


Figure 5. Select Device and Connection Tab

Analog Devices ADF4158/9 PLL Software

File Tools Help

Select Device and Connection: **Main Controls** | Ramps and Shift-keying | Other Functions

RF Settings

RF VCO Output Frequency: MHz

Reference Frequency: MHz

R counter: Ref Doubler: ☐ Ref /2: ☐

PFD Frequency: MHz

Channel spacing: kHz

Prescaler:

INT: + $\frac{\text{FRAC}}{\text{MOD}}$ = MHz

FRAC: MOD:

N = 580

PLL Settings

Register 0: Muxout:

Register 2: CSR: Charge Pump Setting: mA

Register 3: Counter Reset: Charge Pump Tri-State: Phase Detector Polarity: Lock Detect Precision: Power Down: SD Reset: N SEL:

Register 4: CLK Div Mode: FastLock Timer/CLK Div:

Registers

0x 1220000	0x 9	0x 7408002	0x 43	0x 180004	0x 5	0x 6	0x 7
Write R0	Write R1	Write R2	Write R3	Write R4 - Ramp 1	Write R5 - Ramp 1	Write R6 - Ramp 1	Write R7

14:43:19: Firmware loaded.
 14:43:19: USB adapter board connected.
 14:43:24: USB ready.
 14:43:26: USB ready.

ADF4xxx USB Adapter Board connected.

ANALOG DEVICES

0x **800005**

0x **800006**

08972-009

Figure 6. Main Controls Tab

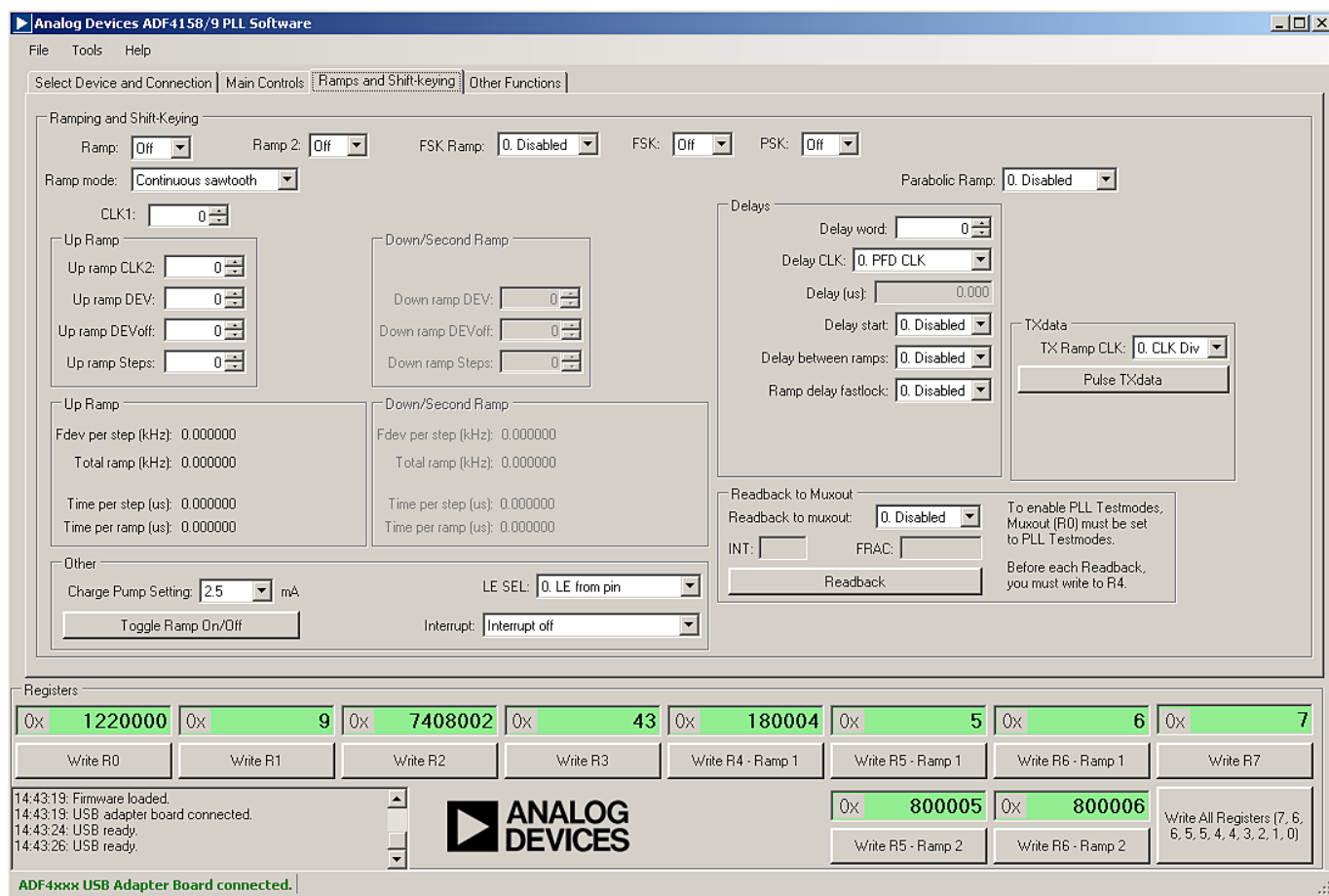
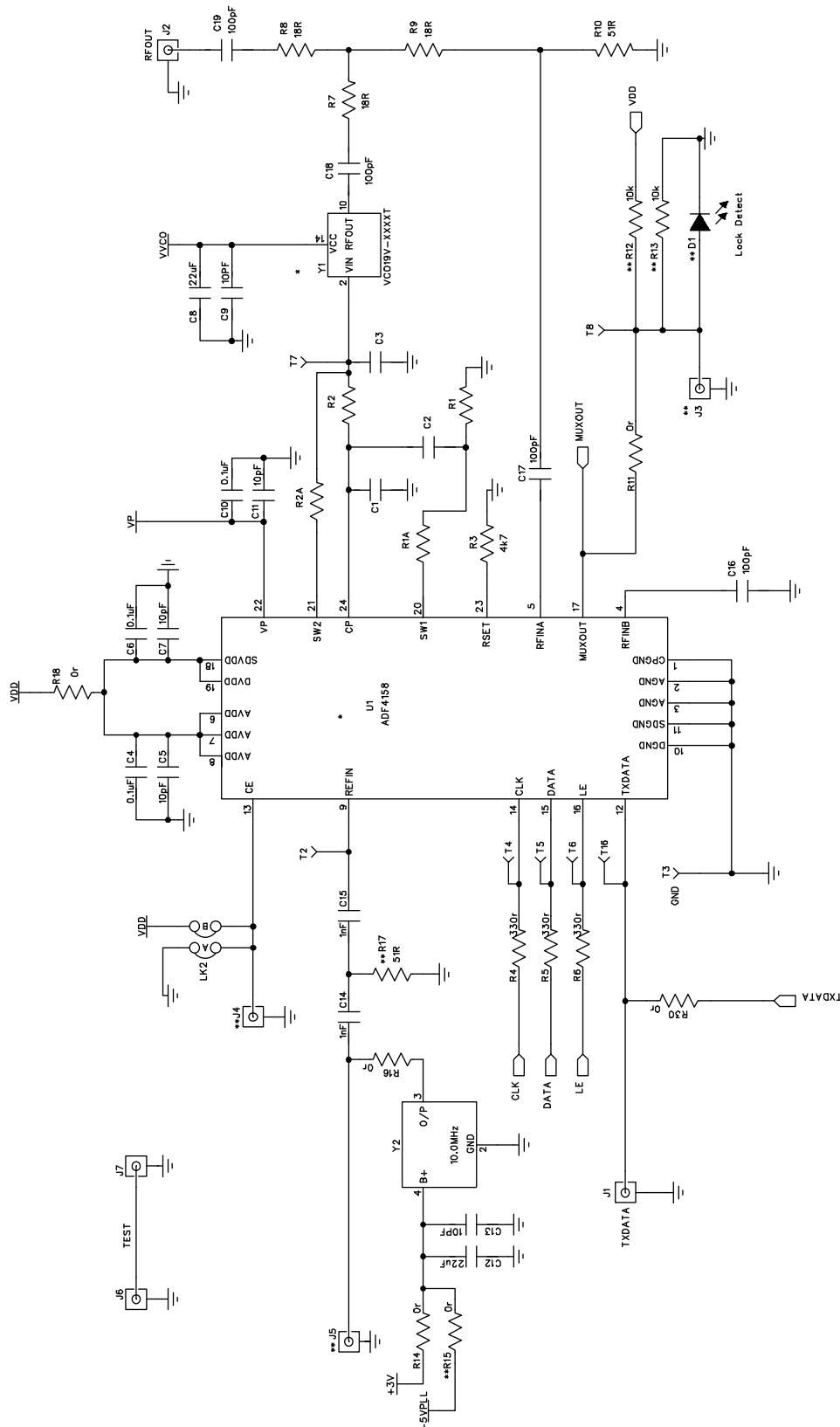


Figure 7. Ramps and Shift-Keying Tab

EVALUATION BOARD SCHEMATIC



Note on non-inserted components.
 ** - These components can be inserted by the user for expansion purposes.

Figure 8. Evaluation Board Schematic (Page 1)

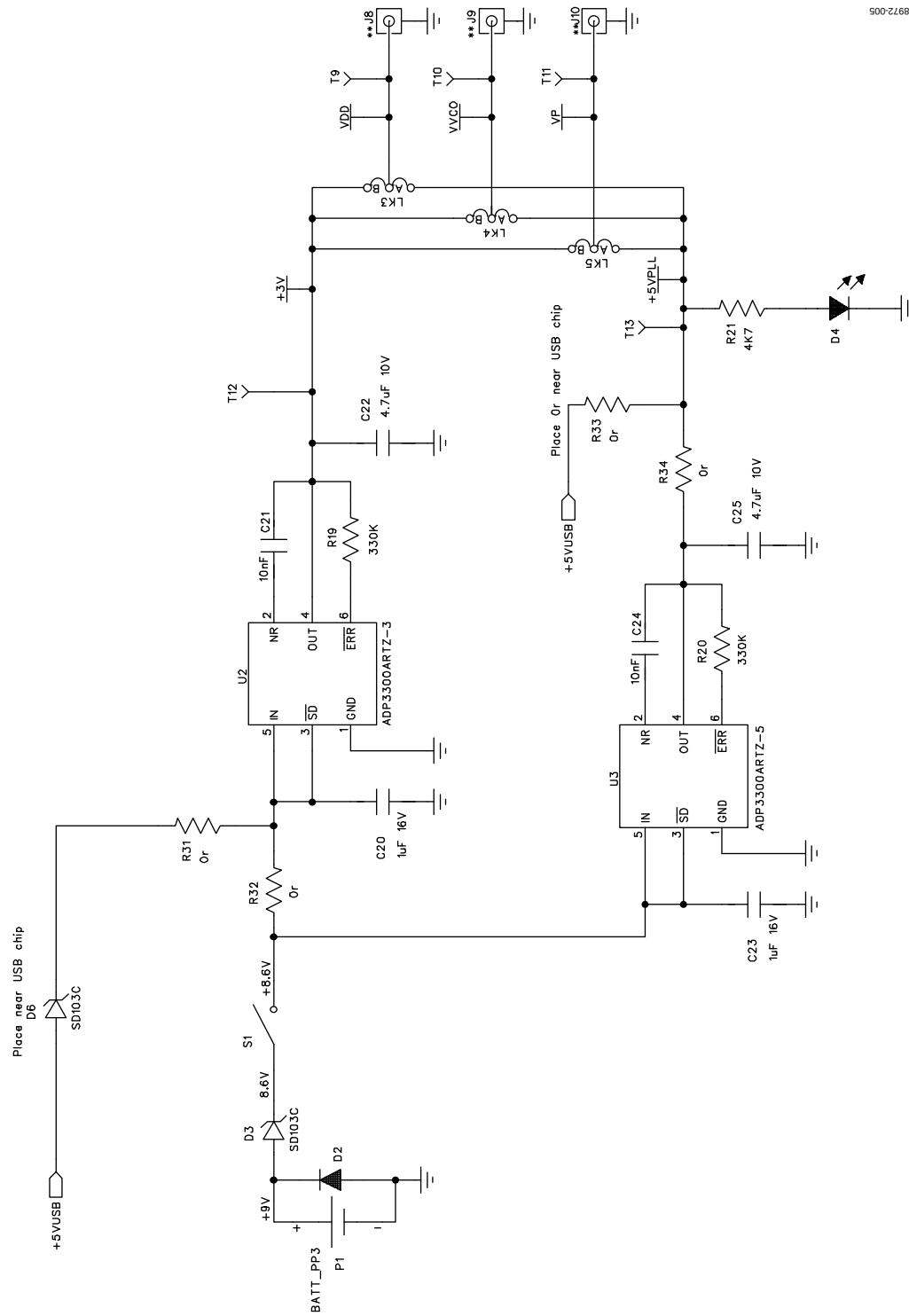
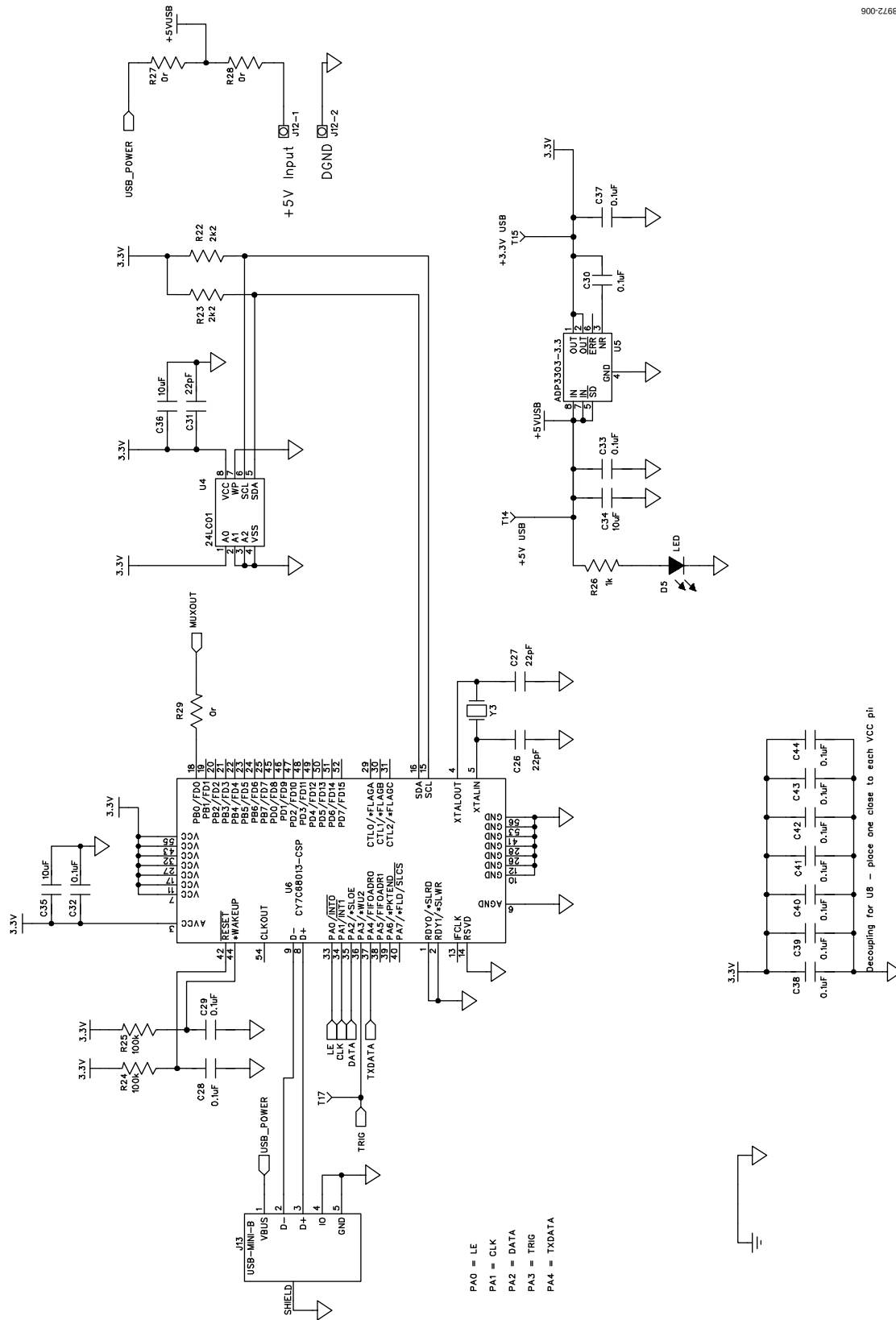


Figure 9. Evaluation Board Schematic (Page 2)



ORDERING INFORMATION

BILL OF MATERIALS

Table 1.

Reference Designator	Description	Manufacturer/Part Number	Supplier Stock Code
C1	Capacitor 680 pF 0603	MCCA000220	FEC 1759083
C2	Capacitor 8.2 nF 0603	MCCA000236	FEC 1759101
C3	Capacitor 330 pF 0603	MCCA000212	FEC 1759075
C4, C6, C10, C28 to C30, C37 to C44	Capacitor 0.1 μ F 0603	CM105X7R104K16AT	FEC 1216538
C5, C7, C9, C11, C13	Capacitor 10 pF 0603	06035A100JAT2A	FEC 499110
C8, C12	Capacitor 22 μ F 0805	GRM21BR60J226ME39L	Digi-Key 490-1719-1-ND
C14, C15	Capacitor 1 nF 0603	06035A102JAT2A	FEC 317202
C16 to C19	Capacitor 100 pF 0603	06035A101JAT2A	FEC 499122
C20, C23	Capacitor 1 μ F 16 V 0603	GRM188R71C105KA12D	Digi-Key 490-3900-1-ND
C21	Capacitor 10 nF 0603	CC0603KRX7R8BB103	FEC 3019561
C22, 25	Capacitor 4.7 μ F 10 V 0603	LMK107BJ475KA-T	Digi-Key 587-1441-1-ND
C24	Capacitor 10 nF 0603	CC0603KRX7R8BB103	FEC 3019561
C26, C27, C31	Capacitor 22 pF 0603	06035A220JAT2A	FEC 498543
C32, C33	Capacitor 0.1 μ F 0603	CM105X7R104K16AT	FEC 1216538
C34 to C36	Capacitor 10 μ F 0805	GRM21BR71A106KE51L	Digi-Key 490-3905-1-ND
D1	LED	DNI	DNI
D2	Diode 1N4001 DO-41	1N4001	FEC 9564993
D3, D6	SD103C 20 V DO35	SD103C-TP	Digi-Key SD103CTPMSCT-ND
D4	Red LED	TLLR4401	FEC 1045504
D5	Green LED 0805	HSMG-C170	FEC 579-0852
J1 to J10	SMA SMA_CARD_EDGE_RF_VIAS	142-0701-851	FEC 1019325
J12	CON\POWER	CTB5000/2	FEC 151789
J13	USB-MINI-B	56579-0576	FEC 9786490
LK2	JUMPER_2_INVTEXT	M20-9980345 and M7567-05	FEC 1022230 and FEC 150410
LK3 to LK5	JUMPER2\SIP3, LINK-3P-NOTEXT	M20-9980345 and M7567-05	FEC 1022248 and FEC 150410
P1	BATT_PP3	593+594	FEC 723988
R1	Resistor 2.4 k Ω 0805	MC 0.1W 0805 1% 2K4	FEC 9332880
R1A	Resistor DNI 0805	DNI	DNI
R2	Resistor 5.1 k Ω 0805	MC 0.1W 0805 1% 5K1	FEC 9333320
R2A	Resistor DNI 0805	DNI	DNI
R3, R21	Resistor 4.7 k Ω 0603	MC 0.063W 0603 4K7	FEC 9331247
R4, R5, R6	Resistor 330 Ω 0603	MC 0.063W 0603 330R	FEC 9331018
R7, R8, R9	Resistor 18 Ω 0603	MC 0.063W 0603 18R	FEC 9330747
R10, R17	Resistor 51 Ω 0603	MC 0.063W 0603 51R	FEC 9331336
R11	Resistor 0 Ω 0603	MC 0.063W 0603 0R	FEC 9331662
R12, R13	Resistor 10 Ω 0603	MC 0.063W 0603 10K	FEC 9330399
R14, R15, R16, R18, R27, R28, R30 to R34	Resistor 0 Ω 0603	MC 0.063W 0603 0R	FEC 9331662
R19, R20	Resistor 330 Ω 0603	MC 0.063W 0603 330K	FEC 9331042
R22, R23	Resistor 2.2 k Ω 0603	MC 0.063W 0603 2K2	FEC 9330810
R24, R25	Resistor 100 Ω 0603	MC 0.063W 0603 100K	FEC 9330402
R26, R29	Resistor 1 Ω 0603	MC 0.063W 0603 1K	FEC 9330380
S1	SW_POWER, SW_SIP-3P	TL36P0050	FEC 1082496
T2, T4 to T17	Test point	20-313137	FEC 8731144
T3	Test point	20-2137	FEC 8731128

Reference Designator	Description	Manufacturer/Part Number	Supplier Stock Code
U1	Fractional-N frequency synthesizer	Analog Devices ADF4158BCPZ	ADF4158BCPZ
U2	Low dropout linear regulator	Analog Devices ADP3300ARTZ-3	ADP3300ARTZ-3
U3	Low drop out linear regulator	Analog Devices ADP3300ARTZ-5	ADP3300ARTZ-5
U4	IC EEPROM 64 kB 400 KHz	Microchip Technologies24LC64-ISN	Digi-Key 24LC64-I/SN-ND
U5	Low drop out linear regulator	Analog Devices ADP3303ARZ-3.3	ADP3303ARZ-3.3
U6	USB microcontroller	Cypress Semiconductor CY7C68013A-56LFXC	Digi-Key 428-1669-ND
Y1	Voltage controlled oscillator	Z Communication VCO V940ME03	VCO V940ME03
Y2	OSC_TCXO 10.0 MHz OSC_TCXO	801-BELF-100	Fox 801-BELF-100 (10 MHz)
Y3	XTAL-CM309S 24 MHz	X24M000000S244	FEC 9509658
	9 V PP3 battery	GP1604-10	FEC 5057050
	Rubber stick-on feet 4x	SJ5076BLACK	FEC 1165061
	Bare PCB	Analog Devices EVAL-ADF4158EB1Z	EVAL-ADF4158EB1Z
	Cable MINI-USB A to MINI-B 1M	Hirose Electric Co., Ltd. UX40-MB-5PA-1000-2003	Digi-Key H2957-ND

**ESD Caution**

ESD (electrostatic discharge) sensitive device. Charged devices and circuit boards can discharge without detection. Although this product features patented or proprietary protection circuitry, damage may occur on devices subjected to high energy ESD. Therefore, proper ESD precautions should be taken to avoid performance degradation or loss of functionality.

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